

Multiple Choice

1. **Answer:** B

Options: A) exactly 3 times longer; B) about 5.2 times longer; C) Not enough information. It will depend on the inclination of the new orbit; D) The length of the year wouldn't change because the Earth's mass stays the same.

Note: Correct option: B - about 5.2 times longer

2. **Answer:** A

Options: A) 10000 times more; B) 100 times more; C) 1000 times more; D) 10 times more

Note: Correct option: A - 10000 times more

3. **Answer:** A

Options: A) The Moon split from the Earth due to tidal forces.; B) The Moon was captured into Earth orbit.; C) The Earth and Moon co-accreted in the solar nebula.; D) Earth was rotating so rapidly that the Moon split from it.

Note: Correct option: A - The Moon split from the Earth due to tidal forces.

4. **Answer:** D

Options: A) Perseus; B) Cygnus; C) Scorpius; D) Leo

Note: Correct option: D - Leo

5. **Answer:** D

Options: A) The Earth must lie completely within the Moon's penumbra.; B) The Moon's penumbra must touch the area where you are located.; C) The Earth must be near aphelion in its orbit of the Sun.; D) The Moon's umbra must touch the area where you are located.

Note: Correct option: D - The Moon's umbra must touch the area where you are located.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'These craters contain the only permanently shadowed regions on Mercury'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Radioactive dating of lunar samples shows that they are older.'

Long Form Response

11. **Answer:** In the inner part of the nebula only metals and rocks were able to condense because of the high temperatures whereas hydrogen compounds although more abundant were only able to condense in the cooler outer regions.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** The luminosity integrated over all wavelengths.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key